

Nick Romano

nicoromano2000@gmail.com • 440-561-7470 • [Portfolio](#), [GitHub](#)

SUMMARY

Engineer-researcher with team-oriented leadership skills as a product-owner in the automation industry, STEAM teaching experience at k-5 and university grade levels, participatory design experience with interactive technologies, and academic research expertise in media studies.

EDUCATION

Parsons School of Design • The New School • New York, NY

Master of Arts in Media Studies • Aug 2024 – May 2026

- **Cumulative GPA:** 4.0/4.0
- **Masters Thesis:** Techno-juggling
- **Awards:** MA, Media Studies Distinguished Thesis Award, May 2026.

Advanced Computing Center for the Arts and Design (ACCAD), The Ohio State University • Columbus, OH • Jan 2024 – May 2024

- **Graduate-Level Coursework:** Designing Experiential Media Systems
- **GPA:** 4.0/4.0

Ira A. Fulton Schools of Engineering • Arizona State University (ASU) • Tempe, AZ

BS in Electrical Engineering • Aug 2018 – Dec 2022

- **Cumulative GPA:** 3.9/4.0
- **Capstone:** A Full-Stack Interface for Residential Solar Energy Viability
- **Awards:** Summa Cum Laude

SKILLS

- **Web Development:** HTML5, CSS3, JavaScript, Responsive Design, Web & Mobile Applications
- **Programming:** Python, API Integration, Source Control (Git), Claude.AI, Electron, NodeJS, A-Frame
- **Data:** Data Visualization (Plotly & Three.js), Data Analysis, REST APIs, AstraDB
- **Other:** Interactive Systems, Soldering, Arduino, HW Testing, Augmented Reality, Projection Mapping, Computer Vision Object Tracking, ProTools, Adobe Premiere, Unity, Blender, MIDI, Touch Designer, Jira, Notion

TEACHING & RESEARCH

Fellow and Technical Lead, Techno-juggling (Interactive Installation) • Ethnographic Media Lab, The New School for Social Research • Cleveland, OH • Oct 2025 – May 2026

- **Showcased Techno-juggling** to a diverse audience of peers and faculty at The New School's Work-in-Process exhibition, New York, NY, April 2026. Educating participants on the mechanics of computer vision and real-time data mapping.
- **Architected a cross-platform interactive environment** using **Electron** and **WebSocket**, enabling real-time spatial data visualization for public educational displays.
- **Managed iterative design cycles** based on documented fieldwork and user testing, translating complex technical feedback into accessible interactive features.
- **Leveraged media archaeology methods** to inform system design, exploring the historical intersections of performance art and emerging technology.

Coding Tutor • The New School • Remote • Jan 2026 – May 2026

- **Facilitated advanced technical practicums for Media Studies students**, translating core programming concepts (HTML5, CSS3, JavaScript) into creative, project-based applications.
- **Mentored student-led prototyping**, providing high-level debugging and architectural guidance for interdisciplinary media installations and web-based art.
- **Developed a scalable e-learning curriculum**, authoring a library of instructional documentation and video tutorials to standardize technical literacy for future academic cohorts.

Project Lead and Grant Recipient • Renaissance Box • Cleveland, OH • Jun 2025 – Sep 2025

- **Project Leadership and Funding:** Secured a **\$1,000 grant** through the Renaissance Box incubator to spearhead the development of large-scale interactive media installations for the **Cleveland Ingenuity Festival** and the **CIA Interactive Media Lab**.
- **Team Management:** Directed a multidisciplinary team including a mixed-media artist and a high school mentee, overseeing the project lifecycle from conceptualization to public exhibition.
- **Iterative Design:** Employed a rigorous R&D workflow featuring **concept mapping, rapid prototyping, and field research**, utilizing community playtesting to refine user UX/UI and hardware reliability.

Key Installations:

- **Collage-Bop:** Developed a computer-vision-based installation where participants manipulate air hockey pucks to generate real-time video collages.
 - Engineered a system using an **overhead camera to track puck coordinates**, mapping spatial data to a user-facing visual interface.
 - Integrated **MIDI keyboard controls** allowing users to modulate audio-visual parameters including volume, geometry, layering, and playback speed.
 - Developed a post-interaction pipeline where users received a unique code to replay their specific creation process via a dedicated web portal.
- **Tell-A-Vision:** Designed and fabricated a robotic human figure inspired by Nam June Paik, constructed from upcycled CRT monitors, telephones, and mobile devices.
 - Programmed limb-mounted monitors to display **dynamic visual content triggered by participant motion**, blending analog aesthetics with modern sensor-based interactivity.

AR Developer, Broadway Theatre's 1920s Marquee Revival • Mount Pleasant, MI • May - Aug 2025

- **Augmented Reality Development:** Engineered a mobile-based WebAR experience designed to drive community engagement and fundraising efforts for the restoration of the historic marquee.
- **3D Modeling and Archival Research:** Conducted extensive historical research using the theatre's physical archives to accurately reconstruct digital assets; utilized **Blender** to model period-accurate 3D marquee components based on original source images and descriptions.
- **Geospatial Implementation:** Traveled on-site to perform **3D Visual Positioning System (VPS) scans** of the theatre facade, ensuring precise digital-to-physical alignment for the AR overlay. **WebAR Deployment:**

Developed and optimized the mobile application using **Niantic 8th Wall**, enabling a seamless, browser-based user experience without the need for external app downloads.

AR Developer, Remixin' • Cleveland Museum of Art Transformer Station • Cleveland, OH • May - Aug 2025

- **Interactive Exhibition Design:** Developed and deployed a custom Augmented Reality (AR) experience exhibited at the **Cleveland Museum of Art's Transformer Station**, bridging physical print media with digital interactivity.
- **WebAR Engineering:** Architected a browser-based application utilizing **Mind.AR, Three.js, and A-Frame** to deliver high-performance camera tracking and real-time 3D overlays without requiring a native app install.
- **AI-Augmented Development:** Leveraged **Claude.AI** as a technical collaborator to accelerate the coding process, optimizing JavaScript frameworks for seamless audio-visual synchronization and interactive link integration.
- **User Experience (UX):** Streamlined audience entry-points via QR-code activation, enabling users to instantly trigger immersive overlays that transformed static prints into dynamic, multi-sensory experiences.

Interactive Design Educator Science, Technology, Engineering, Arts, and Mathematics (STEAM) • Cleveland Play House • Cleveland, OH • May - July 2024

- **Product Engineering & Fabrication:** Designed and constructed "Music Circle," a tactile "musical carpet" interface using vinyl records and conductive hardware integrated with Makey-Makey microcontrollers to facilitate simultaneous play for 8+ students.
- **Software Development:** Engineered a Python-based software suite to drive audio-visual educational games, implementing custom response logic to promote collaborative social learning and digital literacy.
- **Curriculum Leadership:** Developed a research-based STEAM framework and web-enabled instructional environments to deliver immersive tech curricula to K–4 students.
- **Instructional Management:** Led 25-student cohorts through a rigorous 5-week accelerated summer enrichment program, bridging the gap between complex hardware and early-childhood accessibility.
- **Iterative R&D:** Optimized installation engagement and durability through a rapid prototyping lifecycle, utilizing direct student feedback loops and situational design processes.

Interactive Digital Systems Developer • OSU ACCAD • Columbus, OH • Jan – May 2024

- **Researched and engineered interactive environments** using **Isadora**, specializing in real-time projection mapping and sensor-based media triggers.
- **Developed a "Snare-Drum Video Controller"** by interfacing analog percussion with digital outputs via **Makey-Makey**, demonstrating expertise in hardware-software integration.
- **Prototyped a generative audiovisual "Poetry Machine"** and a time-synced Bebop historical narrative, utilizing complex logic to synchronize multi-channel media.
- **Applied iterative design methodologies** under the mentorship of Alex Oliszewski, refining UI/UX principles for immersive, non-traditional performance spaces.

Lead Web Systems Developer, Engineering Capstone • ASU • Tempe, AZ • Jan – Dec 2022

- **Architected a full-stack interactive dashboard** to quantify solar panel viability, translating complex geospatial data into actionable consumer insights.
- **Engineered a multi-source API pipeline** in Python, synthesizing real-time data from Bing Maps, Google Maps, and the NREL (National Renewable Energy Laboratory) database.
- **Developed a high-performance UI** using the Dash (Plotly) framework and custom CSS, ensuring a responsive experience for complex data visualization.
- **Designed a robust relational database schema** to manage and query location-based solar metrics, optimizing data retrieval speeds.
- **Directed the technical lifecycle**, from initial competitive research and UI/UX analysis to final deployment and stakeholder presentation.

Physics I Instructional Assistant • Ira A. Fulton Schools of Engineering • ASU • Tempe, AZ • Aug - Dec 2019

- **Optimized student learning outcomes** for Physics I curriculum through the strategic design of supplemental lesson plans and comprehensive exam prep modules.

- **Facilitated high-impact tutoring sessions** for small cohorts (1–5 students), utilizing targeted intervention strategies to clarify Newtonian mechanics and thermodynamic principles.
- **Synthesized complex theoretical concepts** into digestible instructional assets, including curated lecture notes and visualized problem-solving guides.

PROFESSIONAL EXPERIENCE

Gallery Attendant • Cleveland Museum of Art (CMA) • Cleveland, OH • Oct 2024 – Oct 2025

- **Acted as a technical and cultural translator**, guiding a diverse public through the intersection of traditional art collections and new-media digital exhibitions.
- **Monitored and optimized visitor interaction** within high-traffic zones, maintaining a safe environment while fostering deep engagement with interactive media.
- **Selected by the Cleveland Museum of Art's Exhibition Department for the *CMA Artists at Work* showcase**; developed and deployed a custom Augmented Reality (AR) experience.

Electrical Engineering Product Owner • Rockwell Automation • Cleveland, OH • Jun 2022 – Jan 2024

- **Rapid Career Progression: Promoted to Product Owner** within three months of completing co-op.
- **Led the integration of complex web services**, bridging Jira, Jenkins, and Jama to build a robust analytics infrastructure for organizational reporting.
- **Developed custom automation frameworks** using Python APIs, significantly increasing the efficiency of technical testing and data collection processes.
- **Facilitated cross-functional collaboration**, mentoring and directing a diverse team of 10 engineers in an Agile environment.
- **Designed interactive data visualization dashboards**, translating internal engineering metrics into accessible insights for broad institutional visibility.

Information Management Specialist • Arizona State University • Tempe, AZ • Aug 2019 – Jan 2022

- **Spearheaded the development of data tools focused** on tracking student lifecycle engagement and enhancing program-level education initiatives.
- **Automated departmental communications** utilizing REST APIs, increasing the efficiency of digital content distribution to students and faculty.
- **Leveraged data visualization** to communicate program impact and student outcomes during high-level stakeholder meetings.
- **Managed relational database systems** to streamline the collection and analysis of student metrics, directly supporting institutional reporting requirements.

ENGAGEMENTS

- **Featured Performer & Speaker • Technical Salon and Algorave**, a math-driven generative composition • The Rotunda (UPenn) • Philadelphia, PA • *May 2026*.
 - Synthesize complex code and aesthetic theory into an interactive "salon" format to encourage peer-to-peer technical learning. Showcase technical mastery in mixing diverse media modalities through live algorithmic performance.
- **(Forthcoming) Exhibitor • IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)**, a premier annual computer vision event • Denver, CO • *Jun 2026*.